

Networking-Modules

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TryHackMe Networking Learning

Platform: TryHackMe

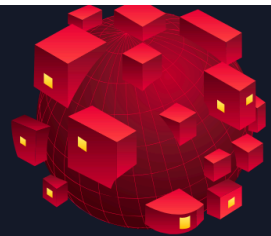
Path: Networking

Completion Status: Completed

Networking

Learn about the OSI model and TCP/IP networking layers. Explore the different plaintext and secure networking protocols that we use every day.

This module aims to introduce you to the ISO OSI and TCP/IP models and present all the main protocols necessary for networks to function. It also introduces all the essential protocols for web browsing, file transfer, email, and remote administration.



under this topic we learned the 2 models of tcp/ip and OSI models

OSI models

7. Application

6. Presentation

5. Session

4. Transport

3. Network

2. Data link

1. Physical

THE PHYSICAL LAYER=1

The physical layer, also referred to as layer 1, deals with the physical connection between devices; this includes the medium, such as a wire, and the definition of the binary digits 0 and 1. Data transmission can be via an electrical, optical, or wireless signal. Consequently, we need data cables or antennas, depending on our physical medium.

THE DATA LINKER LAYER

The Data Link Layer is Layer 2 in the OSI model, responsible for node-to-node data transfer, framing, and error detection. It sits right between the physical hardware layer and the upper network

THE NETWORK LAYER

the network layer handles logical addressing and routing, i.e., finding a path to transfer the network packets between the diverse networks.

THE TRANSPORT LAYER

Layer 4, the transport layer, enables end-to-end communication between running applications on different hosts.

THE SESSION LAYER

The session layer is responsible for establishing, maintaining, and synchronising communication between applications running on different Hosts.

THE PRESENTATION LAYER

The presentation layer ensures the data is delivered in a form the application layer can understand. Layer 6 handles data encoding, compression, and encryption. An example of encoding is character encoding, such as ASCII or Unicode

THE APPLICATION LAYER

The application layer provides network services directly to end-user application

Which layer is responsible for end-to-end communication between running applications?

4

✓ Correct Answer

?

Which layer is responsible for routing packets to the proper network?

3

✓ Correct Answer

?

In the OSI model, which layer is responsible for encoding the application data?

6

✓ Correct Answer

?

Which layer is responsible for transferring data between hosts on the same network segment?

2

✓ Correct Answer

?

On a WiFi, within what will an IP packet be encapsulated?

Frame

✓ Correct Answer

What do you call the UDP data unit that encapsulates the application data?

Datagram

✓ Correct Answer

What do you call the data unit that encapsulates the application data sent over TCP?

Segment

✓ Correct Answer

Use `telnet` to connect to the web server on `MACHINE_IP`. What is the name and version of the HTTP server?

lighttpd/1.4.63

✓ Correct Answer

?

What flag did you get when you viewed the page?

THM{TELNET_MASTER}

✓ Correct Answer

Module 2: Networking Essentials

This module expanded on the basic concepts by introducing how devices identify one another and communicate across different networks.

One of the first topics covered was IP addressing.

Every device connected to a network requires a unique IP address so that it can send and receive information. I learned about both IPv4 and IPv6 addresses, as well as the reasons IPv6 was developed to overcome IPv4 address exhaustion.

Another important concept was subnetting.

Initially, subnet masks seemed confusing, but after several examples, I understood that subnetting allows administrators to divide large networks into smaller, more manageable networks. This improves organization, reduces unnecessary traffic, and enhances security.

The module also explained private and public IP addresses.

Private IP addresses are used inside internal networks and cannot be directly reached from the Internet. Public IP addresses are globally unique and are assigned by Internet Service Providers (ISPs).

Another important concept introduced was Network Address Translation (NAT). NAT allows multiple devices inside a private network to share a single public IP address when accessing the Internet.

The lessons also introduced routing.

Routers determine the best path for packets traveling between different networks. Instead of broadcasting packets everywhere, routers make forwarding decisions based on routing tables.

Understanding routing helped me appreciate how data can travel through multiple intermediate networks before reaching its destination.



Module 3: Networking Core Protocols

This section focused on the protocols that make network communication possible.

ARP (Address Resolution Protocol)

ARP is responsible for mapping IP addresses to MAC addresses inside a local network.

Before this module, I never understood how a computer actually discovered another device's physical address. ARP solves this problem by broadcasting requests asking which device owns a specific IP address.

ICMP

ICMP is primarily used for diagnostics and error reporting.

The most familiar example is the ping command.

When a user runs ping, ICMP Echo Requests are sent to another device, which responds with Echo Replies if reachable.

This protocol is extremely useful for testing connectivity and troubleshooting network issues.

TCP

TCP is a connection-oriented protocol designed to provide reliable communication.

The module explained the famous three-way handshake:

1. SYN
2. SYN-ACK
3. ACK

Only after this handshake is complete does communication begin.

TCP guarantees:

- Ordered delivery
- Error checking
- Retransmission of lost packets
- Reliable communication

Protocols such as HTTP, HTTPS, FTP, SSH, and SMTP all depend on TCP.

UDP

Unlike TCP, UDP is connectionless.

It sends packets without establishing a connection or verifying delivery.

Although less reliable, UDP is much faster because it avoids the overhead of acknowledgements and retransmissions.

Applications such as online gaming, VoIP, live streaming, and DNS commonly use UDP.

Learning the differences between TCP and UDP helped me understand why different applications choose one protocol over the other.

DNS

DNS acts as the Internet's phonebook.

Instead of remembering numerical IP addresses, users simply type domain names.

DNS servers translate these names into IP addresses so browsers know where to send requests.

Without DNS, accessing websites would require remembering IP addresses.

DHCP

DHCP automatically assigns IP addresses to devices joining a network.

Instead of manually configuring every computer, a DHCP server provides:

- IP address
- Subnet mask
- Default gateway
- DNS server

Task 1	✔	Introduction	☰	▼
Task 2	✔	DNS: Remembering Addresses		▼
Task 3	✔	WHOIS		▼
Task 4	✔	HTTP(S): Accessing the Web		▼
Task 5	✔	FTP: Transferring Files		▼
Task 6	✔	SMTP: Sending Email		▼
Task 7	✔	POP3: Receiving Email		▼
Task 8	✔	IMAP: Synchronizing Email		▼
Task 9	✔	Conclusion		▼

Module 4: Networking Secure Protocols

This module introduced protocols designed to protect data during transmission.

HTTPS

HTTP sends data as plain text, making it vulnerable to interception.

HTTPS encrypts communication using TLS, protecting confidentiality and integrity.

Today, nearly every website handling sensitive information uses HTTPS.

TLS

Transport Layer Security provides encryption, authentication, and data integrity.

TLS ensures that:

- Attackers cannot read intercepted traffic.
- Data cannot be modified unnoticed.
- Users communicate with legitimate servers.

Learning about digital certificates and certificate authorities helped me understand how browsers verify website identities.

SSH

SSH provides encrypted remote access to systems.

Unlike Telnet, which sends everything in plain text, SSH encrypts usernames, passwords, and commands.

SSH is widely used by Linux administrators and penetration testers.

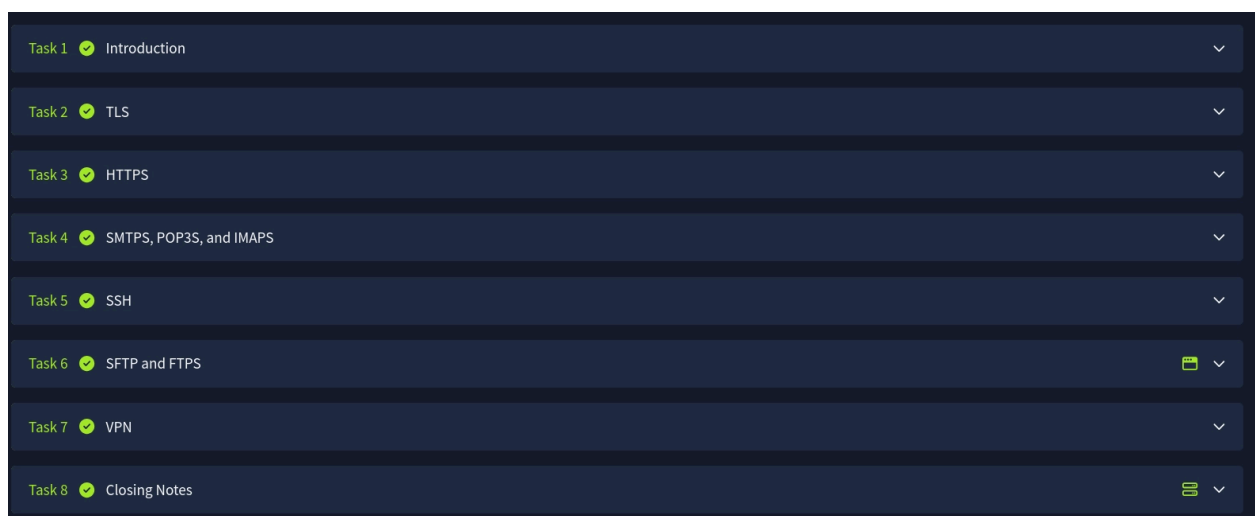
VPN

Virtual Private Networks create encrypted tunnels across public networks.

VPNs provide:

- Confidentiality
- Secure remote access
- Protection on public Wi-Fi
- IP masking

This module clarified how VPNs differ from HTTPS. HTTPS secures individual web sessions, whereas VPNs encrypt all network traffic leaving a device.



Module 5: Wireshark Basics

Wireshark is one of the most powerful packet analysis tools available.

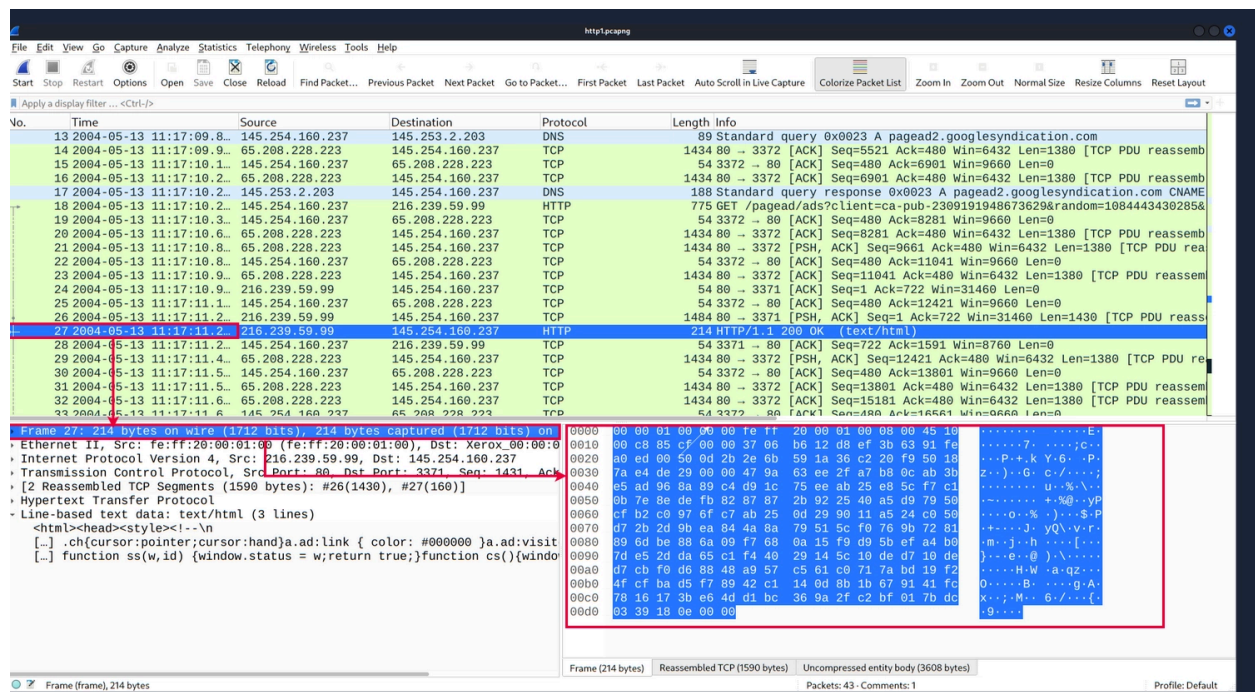
The module taught me how to capture network traffic and inspect packets in detail.

Some useful features included:

- Packet filtering
- Following TCP streams
- Inspecting protocol headers
- Analyzing DNS queries
- Viewing HTTP requests and responses

One of the biggest lessons from Wireshark was realizing how much information is visible when traffic is unencrypted.

By examining packet captures (PCAP files), I became more comfortable understanding how protocols appear "on the wire."



Module 6: Tcpdump Basics

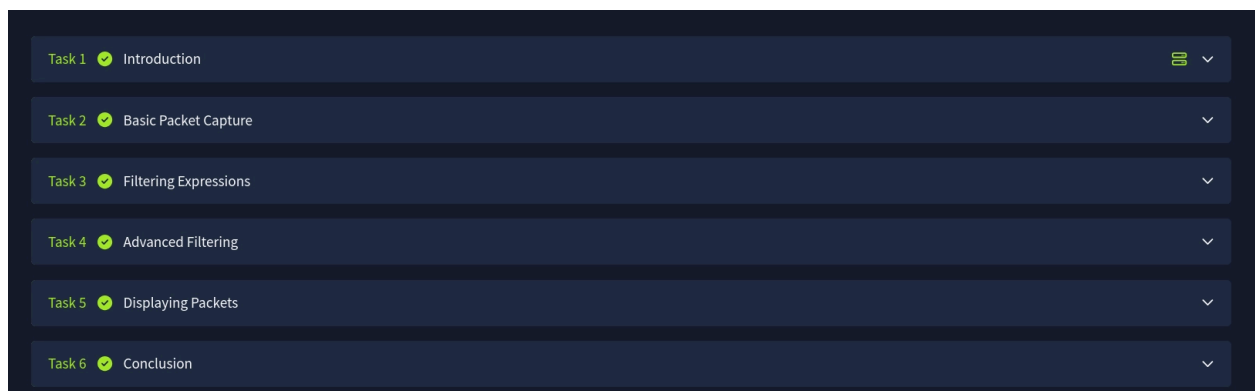
While Wireshark provides a graphical interface, Tcpdump is a command-line packet capture tool.

I learned how to:

- Capture packets
- Save traffic to PCAP files
- Filter by host
- Filter by protocol
- Filter by port
- Analyze captured traffic later

Tcpdump is especially useful on remote Linux servers where graphical interfaces are unavailable.

Learning Tcpdump reinforced the idea that packet analysis can be performed efficiently even without a GUI.



Module 7: Nmap Basics

The final module introduced Nmap, one of the most widely used network reconnaissance tools.

I learned how Nmap helps security professionals discover hosts and services running on a network.

Some important scans included:

- Host discovery
- TCP SYN scan
- TCP Connect scan
- UDP scan
- Version detection
- Operating system detection

I also learned that every open port usually represents a running service.

For example:

- Port 22 → SSH
- Port 80 → HTTP
- Port 443 → HTTPS
- Port 53 → DNS

Understanding ports made me realize how attackers identify potential entry points during penetration testing.

The module also introduced Nmap Scripting Engine (NSE), which extends Nmap's capabilities by allowing vulnerability detection and automated service enumeration.

Practical Skills Gained

Completing this learning path significantly improved my networking knowledge. I can now:

- Explain how the OSI and TCP/IP models work.
- Differentiate between MAC addresses and IP addresses.
- Understand packet encapsulation and transmission.
- Configure and identify subnetting concepts.
- Explain the differences between TCP and UDP.
- Understand how DNS, DHCP, and ARP operate.
- Describe how HTTPS and TLS secure communication.
- Capture and analyze packets using Wireshark.
- Capture traffic using Tcpcat from the command line.
- Perform host discovery and port scanning using Nmap.
- Better understand how penetration testers enumerate network services.

